JSE Digital & Data Academy

The final project

Brief for the 2026 cohort

Issued 7 October 2026

Two years ago, each of you put your name forward for the Academy. You all had your reasons. Some of you were tired of waiting on reports that took days to arrive. Some had watched a process at work break in the same place month after month and wanted to be the person who could fix it. Some were curious about data and technology and wanted to understand what was going on under the surface. A few of you probably just wanted to see what you were capable of.

Since then you have learned how data is managed and governed, how to query and shape it, how good software is designed, how the JSE's markets work from issuance through to settlement, and how to build and test machine learning models. You have presented your work to the people who run those markets and answered their hard questions.

This last project brings all of that back to where you started. We are asking you to go back to the reason you joined, find a real problem in the part of the JSE you know best, and solve it properly, using everything the programme has given you.

The assignment in short

> Working on in a pair, choose a real problem in your business areas. Find someone at the JSE who owns that problem and wants it solved. Use any data, people and approved tools you can get access to. Build something that works, show that it makes a measurable difference, and present it on Friday 27 November 2026.

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1 Start with your reason

Before you choose a problem, write a short note, no more than a page, about why you joined the Academy. Be specific. These questions may help:

- What was frustrating you at work at the time you applied?
- Was there a task you did over and over that you knew could be done better?
- What did you hope you would be able to do by the end?
- Looking back, which part of the programme changed how you see your own work?

This note is part of your submission, and it will be read. It also helps you choose. The best projects tend to come from something that has bothered you for a long time.

2 Choose your problem

Pick a problem from the business area you work in or know well. It should be a problem that people at the JSE actually have today, not one invented for the project. A good problem usually meets all of the following:

- Someone at the JSE owns it and would be glad to see it solved.
- You can describe how it is handled today, and roughly how long or how much it costs.
- The data you need exists and you have a realistic chance of getting access to it in time.
- You can make a real difference to it in about six weeks, even if the full solution would take longer.
- You can measure whether your solution helped.

Some examples of the kind of problems we mean, by business area:

Business area	Examples
Marketing	Briefing packs for client meetings built from real trading and data-usage records. Measuring which campaigns and events lead to new data subscriptions or listings.
Finance	The reconciliation use case (see below). Checking invoices against contracted rates. A tracker that shows what the JSE's AI initiatives actually cost and save.
Capital Markets	Flagging closing prices likely to move sharply the next morning. Spotting unusual patterns in order data for the surveillance team to review. Speeding up the work involved in launching a new derivative contract.
Technology & Information Services	Measuring how much AI coding tools actually speed up delivery on a real project. Checking the accuracy of data extracted from SENS announcements. A single reliable view of instrument reference data.

If you can't find a problem of your own, or you would rather take on one that is already scoped, you can choose one of these:

- The finance reconciliation use case, with Lesedi's team as sponsor. Its business rules are being validated by Finance now, and Finance candidates get first choice.

- Any of the seven use cases in the cohort use-case portfolio, such as contract-versus-invoice reconciliation or the value-realisation tracker.

Choosing a ready-made problem is perfectly acceptable. You still need to write your motivation note and explain why this problem matters to you.

3 Find a sponsor and get access

Every project needs a sponsor: a person at the JSE who owns the problem, will give you some of their time, and will tell you at the end whether your work helped them. Your line manager can be your sponsor if the problem is in their area. A head of department is better if the problem crosses teams.

This is your last project, and you may use anything at your disposal to make it succeed. In practice that means:

Data. Any JSE data you can get approved access to. Ask early, because access requests take time. The Academy office will help you write the request and chase it.

People. Any subject matter expert at the JSE who is willing to help. Desk heads, Finance, Market Operations, Surveillance, Information Services, Technology. Most people are happy to give an hour to someone who has done their homework. The Academy office can make introductions.

Tools. Anything on the JSE's approved platforms: Azure, including AI Foundry and Data Lake, Microsoft Fabric, Power BI, the approved AWS services, GitHub Copilot and Microsoft Copilot. If you need something that isn't on the list, ask before you use it.

The Academy. Weekly office hours, the module facilitators, and the work of earlier cohorts.

Get your sponsor to sign off a one-page problem statement before you start building. It saves a lot of trouble later.

4 Build it and measure it

Your project should draw on at least three areas of the programme. You don't have to use machine learning. If a well-designed query and a clear dashboard solve the problem, that is a better answer than a model nobody needs. What we want to see is that you chose the right tools for the problem and can explain why.

Area of the programme	What we would expect to see
Data management and governance	You know where your data comes from, who owns it, how good it is, and you have handled it within policy.
SQL and data engineering	Your data preparation can be run again and gives the same result.
Software design	Your solution is built in parts that make sense, and someone else could maintain it.
Capital markets	You understand the business process behind your data and can explain it to the people who run it.

Area of the programme	What we would expect to see
Machine learning and AI	If you used a model, you compared it with a simple alternative and tested it on data it hadn't seen.
Presenting with data	Your charts and your story make the point clearly to someone outside your field.

Measure how things are done now before you change anything. Record how long the task takes, how often it goes wrong, or what it costs. Then measure the same things with your solution in place. Without the before-and-after comparison, it is very difficult to show that your work made a difference.

Count your costs as well as your benefits: your time, any cloud or licence costs, and the time your sponsor and their team spent helping you. This is the same test the COO applies to every AI initiative at the JSE, and it is reasonable to apply it here.

5 Tell the story

Your final presentation is 15 minutes, followed by 10 minutes of questions. We suggest you follow this storyline, although you can adapt it:

1. Why I came. Two minutes on why you joined the Academy, taken from your motivation note.
2. The problem. What happens today, who it affects, and what it costs.
3. What I found. What you learned from the data and from the people involved. This is often where the surprises are.
4. What I built. A short demonstration, using real data if possible.
5. What changed. Your before-and-after measurements, including the costs.
6. What happens next. Who will keep using it, what it would take to put it into production, and what you would do with more time.

The panel will ask questions like the ones raised during this year's presentations:

- What would happen if you ran the simplest possible approach instead? How much better is yours?
- If you ran your data preparation again tomorrow, would you get exactly the same result?
- Which of your numbers did you check by hand?
- Who will use this on Monday morning, and what will they do differently?
- What does it cost to run, and is it worth it?

6 Working alone or in a pair

You will work in pairs. A variety of factors were considered in the pairing: skills diversity, strengths and it is key that you encourage each other and make it work, as you have throughout the programme. Furthermore, pairs from different business areas often produce the strongest work. The expectation:

- Each of you writes your own motivation note.
- You agree at the start who is responsible for which part of the work, and you write it down.

- Both of you present, and both of you answer questions.
- You are assessed individually as well as together, so each of you must be able to explain the whole solution.

7 What you hand in

Deliverable	Due	Detail
Motivation note	Friday 16 October	One page. Why you joined, and why this problem.
Problem statement	Friday 23 October	One page, signed off by your sponsor. Includes your data and access plan.
Mid-point check-in	Friday 6 November	Fifteen minutes with the Academy team. Show your baseline measurements and early results.
Working solution	Friday 20 November	Code, model, dashboard or process, in a shared repository or workspace, with instructions to run it.
Written report	Friday 20 November	No more than six pages: the problem, what you did, what you measured, what it cost, and what you learned.
Final presentation	Friday 27 November	Fifteen minutes plus ten minutes of questions.
Handover note	Friday 27 November	One page for your sponsor on how to keep using the solution.

8 How it will be assessed

Criterion	Weight	What the panel will look for
Problem and motivation	10%	A real problem that matters to the business, and a clear link to why you joined.
Understanding of the business	15%	You understand the process and the people, and your sponsor recognises their work in your description.
Use of data	15%	Appropriate data, handled correctly, with its quality checked and its limits explained.
Quality of the solution	25%	It works, it suits the problem, and it was built with care.
Evidence of impact	20%	Before-and-after measurements, costs included, and a comparison with a simpler approach.
The story	10%	A clear presentation and good answers to the panel's questions.
Handover	5%	Your sponsor can keep using it without you.

The panel will include your sponsor, a member of the Academy team, and a senior leader from outside your business area. To pass, you need at least 50 percent overall, and your sponsor must confirm that your work helps with the problem you set out to solve.

9 Ground rules

- Use real data where you can. If you have to use sample or synthetic data, say so clearly on every slide where it appears.
- Follow the JSE's data policies. Remove client, supplier and staff names unless you have explicit permission to use them.
- Do not change any production system. If your solution would need to go live, describe what that would take, including the ADA, security and legal reviews.
- Nothing you build should sit in the path of price formation, trading, clearing or settlement.
- You may use AI tools such as Copilot to help you write code or documents. Say where you used them and make sure you understand and can explain everything you submit.
- If you get stuck on access or data, tell the Academy office within a week, not at the end.

10 Timeline and support

Date	Milestone
Wednesday 7 October	Brief issued
Friday 16 October	Motivation note in.
Friday 23 October	Problem statement signed off by your sponsor. Data requests submitted
26 October to 13 November	Build and measure. Office hours every Friday
Friday 6 November	Mid-point check-in
Friday 20 November	Solution and report submitted. Dry runs
Friday 27 November	Final presentations at One Exchange Square

Questions about the brief, sponsors or access go to Porsha in the Academy office. For anything else, speak to me directly.

We are looking forward to seeing what you choose.

Tshivhidzo Makungo, Head of Software Engineering and Academy sponsor